

Class-2: Phasmidia

In the members of this class amphids are simple and **phasmids** are present. Excretory system is well developed.

Example: *Ascaris* (roundworm), *Ancylostoma* (hookworm), *Enterobius* (pinworm), *Wuchereria* (filarial worm).

3.6 PHYLUM-ANNELIDA

-Segmented body plan

3.6.1 General Characters

The term annelida (L. *annulus* : little ring and Gr. *edios* : form) was coined by **Lamarck**. They may be aquatic (marine and freshwater) or terrestrial; free-living, and sometimes parasitic. They exhibit organ-system level of body organisation and bilateral symmetry. They are triploblastic, **metamerically segmented (metamerism)** and **coelomate (schizocoelic coelom)** animals (Fig. 3.10-3.12). Their body and the coelom are divided by transverse septa into segments or **metameres**. They possess longitudinal and circular muscles in the body wall, which help in locomotion. Coelom with coelomic fluid provides a **hydrostatic skeleton** giving stiffness to the body parts thus aiding in locomotion. Aquatic annelids such as *Nereis* possess lateral appendages, the **parapodia**, which help in swimming; chitinous *setae* and *suckers* also help in locomotion in some. Cephalization is more pronounced with distinct head and sense organs in some polychaetes. Respiration is by simple diffusion through the body wall and in *Nereis* highly vascularised parapodia help in respiration. A *closed* circulatory system is present. Respiratory pigments such as **haemoglobin**, **chlorocruorin** are found dissolved in the plasma (a feature unique to the invertebrates). Nephridia (metanephridia- nephridia with nephrostomes), which are 'ectodermal' in origin, help in osmoregulation and excretion. Nervous system consists of paired (cerebral and sub-pharyngeal) ganglia connected by lateral nerves around the pharynx forming a **nerve ring** and to a ganglionated 'double ventral nerve cord' (annelidan type of nervous system). Sense organs like eyes, palps and tentacles occur in some. They may be dioecious (sexes are separate-e.g. *Nereis*) or monoecious (hermaphrodites – e.g. earthworms and leeches). Reproduction is usually sexual. Cleavage is holoblastic (complete) and spiral. Development is direct in the monoecious (bisexual) annelids and indirect in the dioecious (unisexual) annelids. Characteristic larval form, the **trochophore** occurs in the life history.

Examples: *Nereis*, *Pheretima*, *Megascolex* and *Hirudinaria*.

3.6.2 Classification of the phylum Annelida

Phylum Annelida is classified into three classes, namely

1. Polychaeta 2. Oligochaeta 3. Hirudinea

Class-1: Polychaeta

Polychaeta (*poly* : many; *chaetae* : setae) includes marine annelids commonly called bristle worms. Some are free-moving and others burrowing or tubicolous (tube dwelling). Head is distinct with sense organs such as eyes, tentacles and palps. **Parapodia** which bear many setae (hence the name polychaeta) help in locomotion and respiration. Clitellum is absent. They are dioecious (unisexual); gonoducts are absent; gametes are shed into the coelom and passed out through the nephridiopores. Fertilization is external; development includes a trochophore larva.

Examples: *Nereis* (sandworm or ragworm or clam worm), *Aphrodite* (sea mouse), *Arenicola* (lugworm).

Class-2: Oligochaeta

Oligochaeta (*oligos* : few; *chaetae* : setae) includes earthworms which live in moist soil and some live in freshwater (e.g. *Tubifex*). They lack distinct head. Chitinous setae help in locomotion. Clitellum forms a cocoon during the breeding season. They are monoecious (bisexual); development is direct without larval stage.

Examples: *Pheretima*, *Megascolex* and *Tubifex*.

Class-3: Hirudinea

Hirudinea (*hirudo* : leech) includes *leeches*; all are ectoparasites; majority live in fresh water; some are marine and others live on moist land (terrestrial). They have dorso-ventrally flattened body with a **definite number** of segments ; the segments are externally subdivided into **annuli**; internal segmentation is absent ; suckers help in locomotion. Clitellum is conspicuous during the breeding season only. Coelom is filled with a characteristic tissue called **botryoidal tissue** (*botryoidal*: resembling a bunch of grapes; opinions differ on the function of botryoidal tissue - they range from excretion to storage of iron, calcium, revascularization in areas of injury etc.). They are monoecious (hermaphroditic); male reproductive system possesses a copulatory organ, the *cirrus*. Fertilization is internal and development is direct .

Examples: *Hirudinaria* (freshwater leech), *Pontobdella* (marine leech), *Haemadipsa* (land leech).

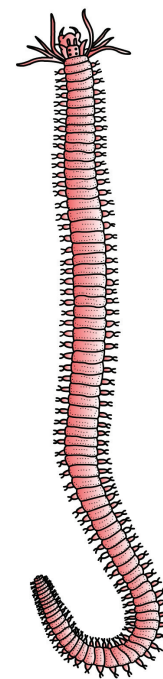


Fig. 3.10 Nereis



Fig. 3.11 Leech

3.6.3 EARTHWORM



Fig. 3.12 Earthworm

Earthworm is a reddish brown terrestrial invertebrate that inhabits the upper layers of the moist soil rich in decaying organic matter. It is nocturnal and during the day time, it lives in burrows made by boring and swallowing the soil. In gardens, they can be traced by their faecal deposits known as **worm castings**. They are considered, the **Friends of farmers**. The common Indian earthworms are *Pheretima*, *Lumbricus* and *Megascolex*. *Drawida grandis* of south India is the longest earthworm in India. *Chaetogaster annandalei* is the smallest earthworm, while *Megascolides australis* is the longest earthworm.

I. Morphology

Pheretima posthuma has a long cylindrical body which is divided into 100 to 120 segments (metameres). The dorsal surface of the body is marked by a dark mid-dorsal line (representing the dorsal blood vessel) along the longitudinal axis of the body. The ventral surface of the body shows genital apertures (pores). The anterior end bears a

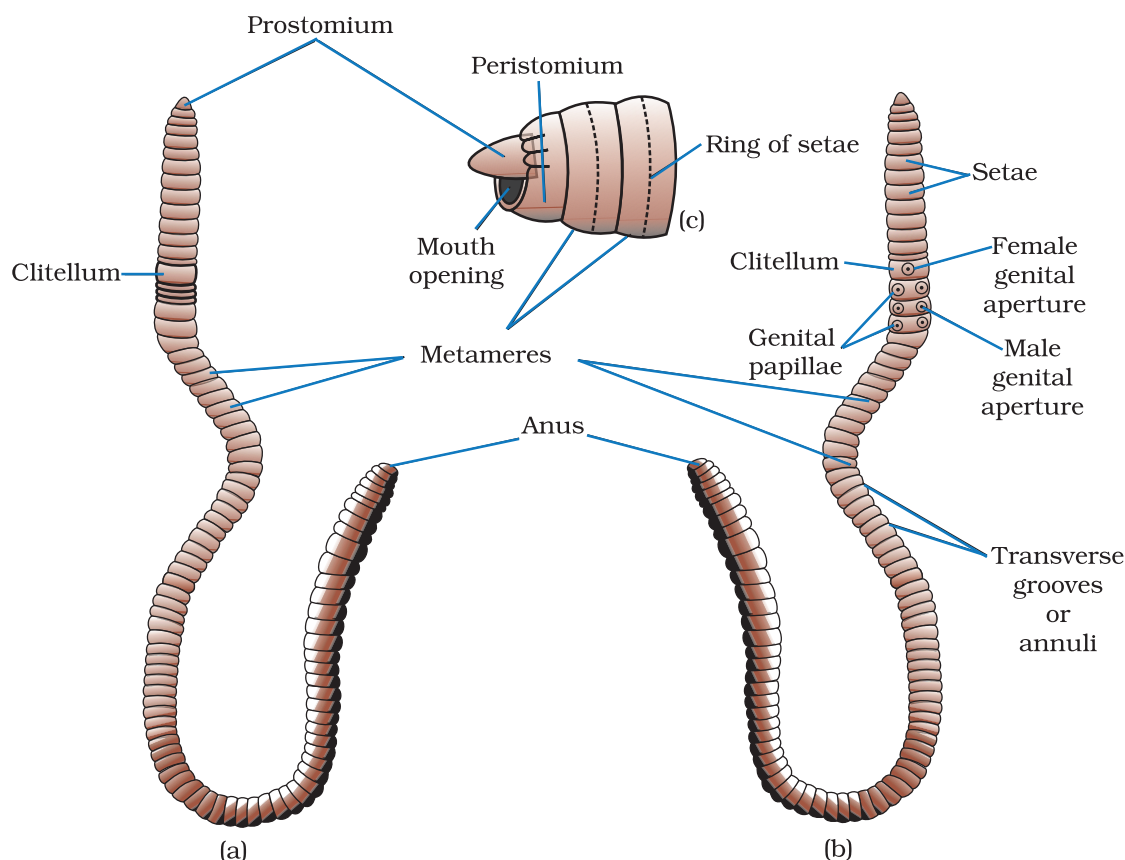


Fig. 3.13 Body of earthworm: (a) dorsal view (b) ventral view (c) lateral view showing mouth opening

lobe which serves as a covering for the mouth, the **prostomium**, which is not a true segment and over hangs the oral aperture. The prostomium bears some sense receptors. The first body segment is the **peristomium** (*buccal segment*) which surrounds the mouth and the prostomium. In a mature worm, the segments 14–16 are covered by a prominent dark band of glandular tissue called **clitellum** or **cingulum** (*L. clitellum*: a pack saddle; *cingulum*: a belt). The body is divisible into three prominent regions –the **pre-clitellar**, **clitellar** and **post-clitellar** regions (Figure 3.13). Four pairs of spermathecal apertures are situated on the ventro-lateral sides of the inter-segmental grooves, from 5th to 9th segments (5/6, 6/7, 7/8 and 8/9). A single female genital pore is present in the mid-ventral line of the 14th segment. A pair of male genital pores is present on the ventro-lateral sides of the 18th segment. Numerous minute pores called nephridiopores open on the surface of the body from the 3rd segment to the last. In each body segment, except the first, last (pygidium) and clitellum (in mature worms) there are rows of s-shaped chitinous **setae** or **chaetae** embedded in the epidermal pits (setal or setigerous sacs) in the middle of each segment. Setae can be protruded or retracted. They play an important role in locomotion.

II. Anatomy

The body wall of earthworm is covered externally by a thin non-cellular cuticle below which is the epidermis, two muscle layers (outer circular and inner longitudinal) and the innermost layer, the outer coelomic epithelium (parietal peritoneum). The epidermis is made up of a single layer of columnar epithelial cells which contain secretory gland cells. The coelomic epithelium is derived from the mesoderm and consists of the outer parietal (somatic) layer and inner splanchnic (visceral) layer. The space between the body wall and alimentary canal is called coelom (perivisceral cavity). It is a *schizocoelom* and is divided into compartments by septa filled with coelomic fluid. The coelomic fluid provides a '**hydrostatic skeleton**' for the earthworm during locomotion.

III. Digestive system

The alimentary canal is a straight tube and runs from the first to the last segment of the body (Figure 3.14). The mouth opens into the **buccal cavity** (1–3 segments) which leads into the muscular **pharynx** (4th segment). A small narrow tube, **oesophagus** (5–7 segments), continues into a muscular **gizzard** (8th segment). It helps in grinding the small particles of food the decaying leaves (**grinding mill**).

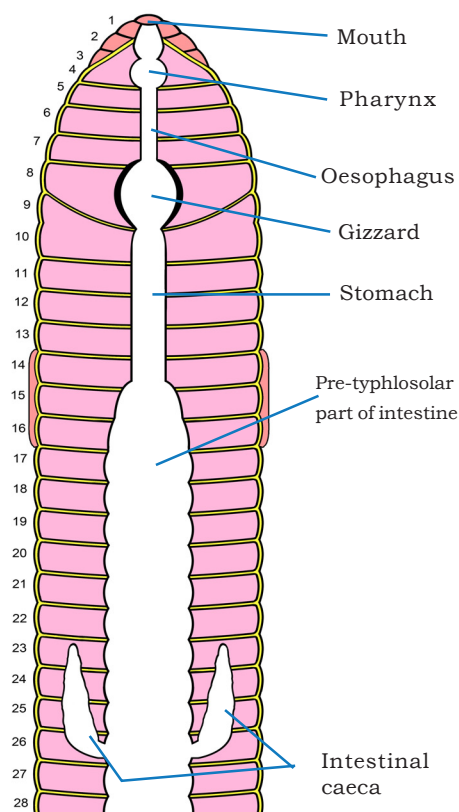


Fig. 3.14 Alimentary canal of earthworm

The stomach extends from the segments 9 to 14. The food of earthworm is decaying leaves and other organic matter mixed with the soil. Calciferous glands, present in the **stomach**, neutralise the humic acid present in the humus of the soil. The intestine starts from the 15th segment and continues till the last segment. A pair of short and conical intestinal caecae project from the intestine in the 26th segment. An internal median fold of the dorsal wall of the intestine called **typhlosole**, helping in increasing the area of absorption, is poorly developed in *Pheretima* (between the 26th and the **rectum**, which occupies the last 23 to 28 segments). The alimentary canal opens to the exterior by a small rounded aperture called anus. The ingested soil rich in organic matter passes through the digestive tract where digestive enzymes breakdown complex food into smaller absorbable units. These simpler molecules are absorbed through intestinal membranes and are utilised for various metabolic activities.

IV. Blood vascular system

Pheretima exhibits a **closed type** of blood vascular system, consisting of blood vessels, capillaries and 4 pairs of hearts. (Figure 3.15). Hearts connect the dorsal and ventral blood vessels. Due to closed circulatory system, blood is confined to the blood vessels. Contractions of the DBV keep blood circulating in the body. Smaller blood vessels supply blood to the gut, nerve cord and the body wall. **Blood glands** are present in the 4th, 5th and 6th segments. They produce blood cells and 'haemoglobin' which is 'dissolved in the plasma'. Blood cells are phagocytic in nature.

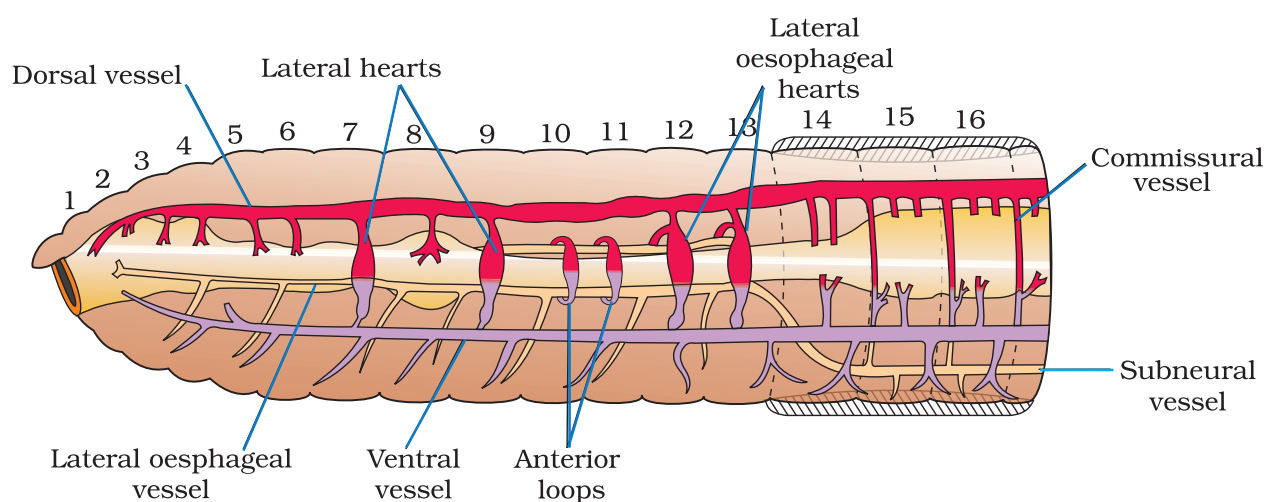


Fig. 3.15 Earthworm circulatory system

V. Respiratory system

Earthworms lack specialised respiratory structures. Respiratory exchange of gases occurs through the moist body surface by diffusion.

VI. Excretory system

The excretory organs occur as segmentally arranged coiled tubules called **nephridia** (sing: *nephridium*). They are of three types:

i) septal nephridia, present on both the sides of the inter-segmental septa of segments 15th/16th to the last. They open into the intestine

ii) integumentary nephridia, attached to the inner body wall from the 3rd segment to the last. They open to the exterior on the body surface by nephridiopores and

iii) pharyngeal nephridia, present as three paired tufts in the segments 4th, 5th and 6th (Figure 3.16). They open into the buccal cavity and pharynx. The different types of nephridia of *Pheretima* are fundamentally similar in structure. Nephridia with nephrostomes are called **open** nephridia (e.g. septal nephridia) and the nephridia which do not have nephrostomes are called **closed** nephridia (e.g. pharyngeal nephridia and integumentary nephridia). The nephridia that open to the exterior through nephridiopores are called '**exonephric nephridia**' and those opening into the gut are called '**enteronephric nephridia**'. Enteronephric nephridia have a role in **osmoregulation**/conservation/**homeostasis** of water. Earthworms are mostly *ureotelic* animals. The chief nitrogenous excretory waste is urea.

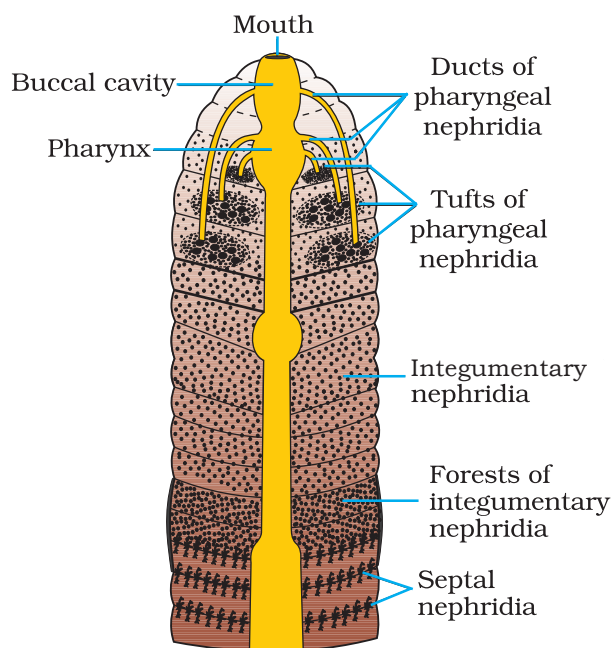


Fig. 3.16 Nephridial system in earthworm

VII. Nervous system

Nervous system consists of a **nerve ring** around the pharynx (in the 3rd and 4th segments) and a double ventral nerve cord. The cerebral ganglia (brain) along with other nerves in the ring integrate sensory input as well as command muscular responses of the body. Sensory system consists of receptor cells that receive light, touch and chemical stimuli. These sense organs are located mostly in the anterior part of the body.

VIII. Reproductive system

Earthworm is a hermaphrodite (bisexual) (Figure 3.17). There are two pairs of **testes** one pair each present in the 10th and 11th segments. Their **vasa deferentia** run up to

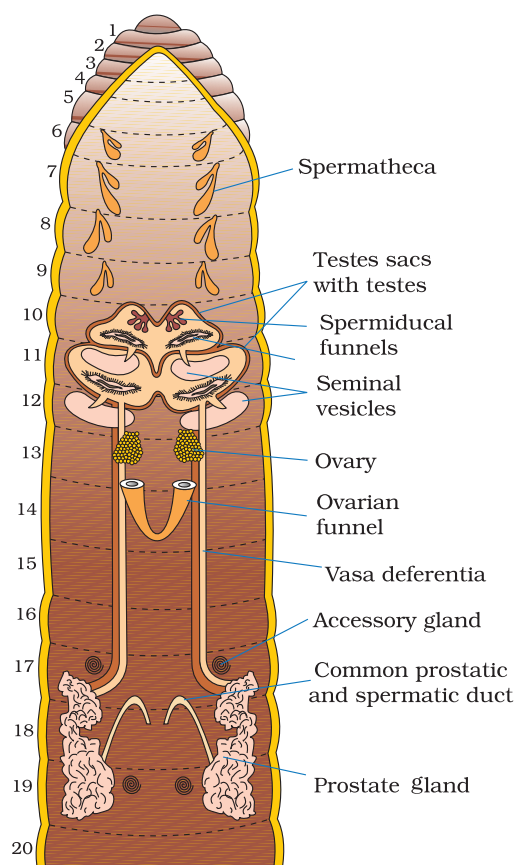


Fig. 3.17 Reproductive system of *Pheretima*

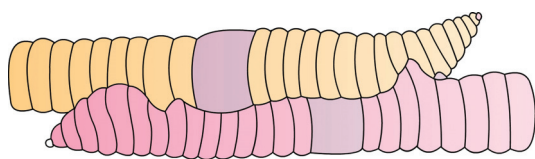


Fig. 3.18 Copulation

the 18th segment where they join the prostatic ducts. Two pairs of seminal vesicles present in the 11th and 12th segments are sacs in which spermatogonia mature into spermatozoa. The common prostatic and spermatic ducts open to the exterior by a pair of male genital pores on the ventro-lateral sides of the 18th segment. Two pairs of accessory glands are present one pair each in the 17th and 19th segments. Four pairs of **spermathecae** are located in the segments 6th to 9th (one pair in each segment). They receive and store spermatozoa (**spermatophore**) during copulation.

One pair of ovaries is attached to the posterior face of the inter-segmental septum of the 12th and 13th segments. *Oviducal funnels* are present beneath the ovaries and they continue into oviducts (14th segment). They join together and open to the exterior on the ventral side of the 14th segment by a single median female genital pore.

IX. Fertilization, cocoon formation and Copulation

Earthworm is protandrous. Mutual exchange of spermatozoa occurs between two mature worms during mating. They mate juxtaposing opposite gonadal openings, exchanging packets of sperms called **spermatophores** (Ref. NCERT book). Mature sperms and egg cells and nutritive fluid produced by the gland cells of the clitellum are deposited in **cocoons**. Fertilisation and development occur within the cocoons which are deposited in the moist soil. The cocoon contains the developing embryos. After about 3 weeks, each cocoon produces on average four earthworms (Ref: NCERT text book). Development is direct i.e., there is no larval stage.

X. Economic importance

Earthworms are known as **Friends of farmers** because they make burrows in the soil and make it porous, which helps in aeration of the soil. The process of enriching the fertility of soil by earthworms involves the process called **vermi-composting**. Earthworms are also used as bait in fishing.